

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of canopy height per plot (esy4 = 4qm plots)

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```
library(DHARMA)
library(emmeans)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.height) %>%
filter(y < 1) # see section Outliers: Exclude site N_DAM (more or less only the tall grass Arrhenatherum)
```

Load data

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `esy4 = fct_relevel(esy4, "R", "R22", "R1A")`.
## Caused by warning:
## ! 1 unknown level in `f`: R
```

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.4634648 0.4485813 0.4336978
```

```
median(sites$y)
```

```
## [1] 0.44
```

```
sd(sites$y)
```

```
## [1] 0.1285512
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 0.228 0.686
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      101
## 2 664       89
## 3 686       99
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord> <int>
## 1 positive      45
## 2 restored     219
## 3 negative      25
```

```
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22     208
## 2 R1A      81
```

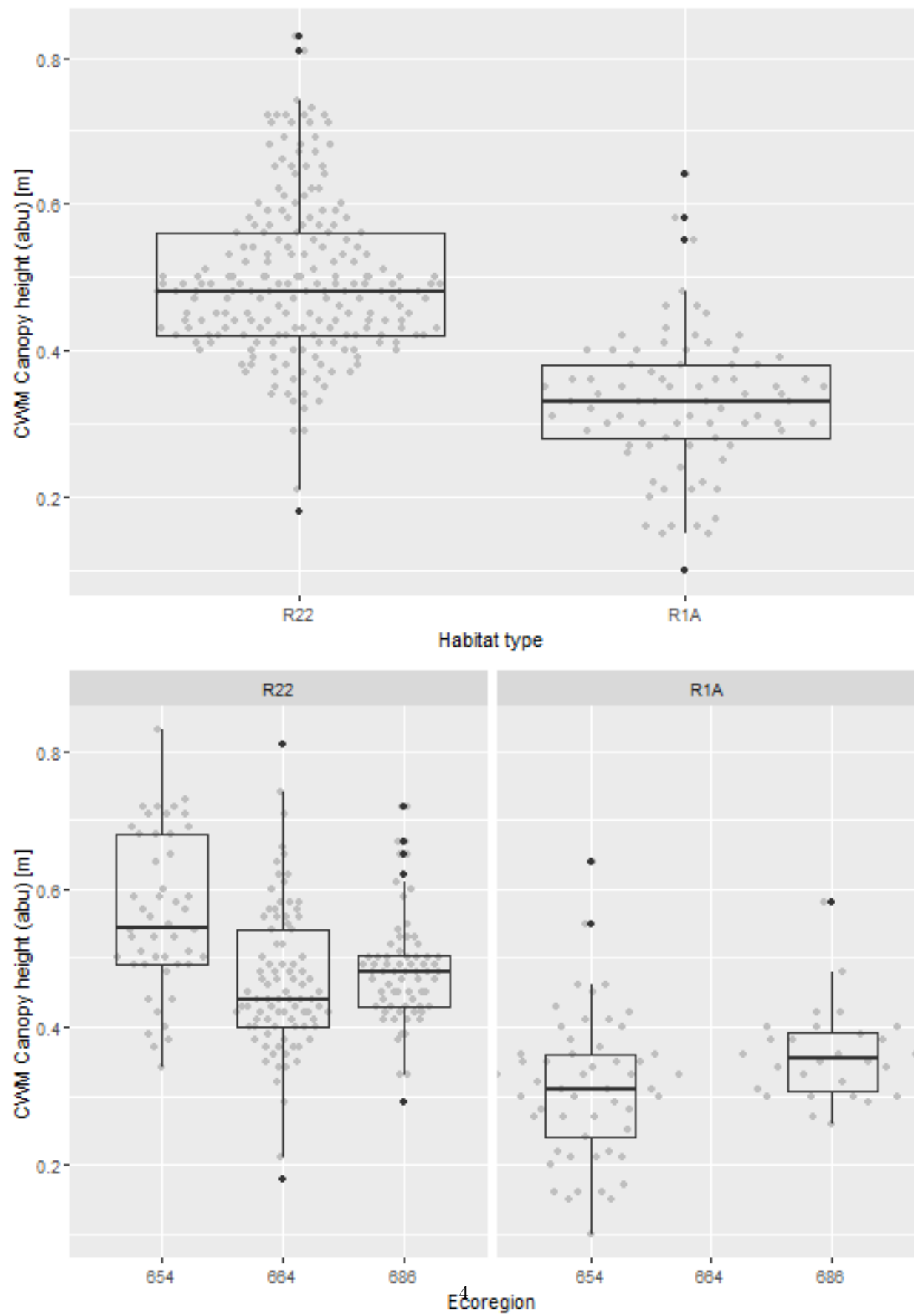
```
sites %>% count(esy4, eco.id)
```

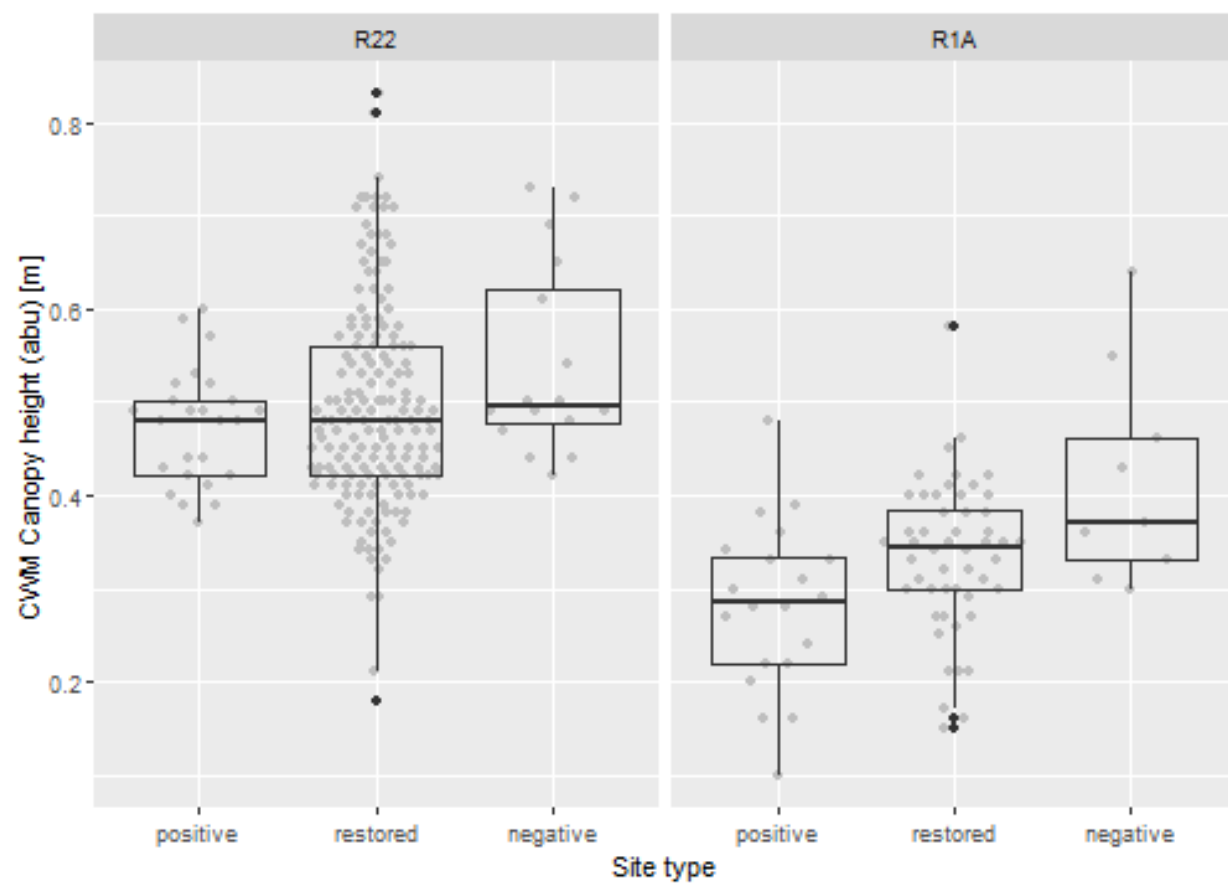
```
## # A tibble: 5 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R22  654      48
## 2 R22  664      89
## 3 R22  686      71
## 4 R1A  654      53
## 5 R1A  686      28
```

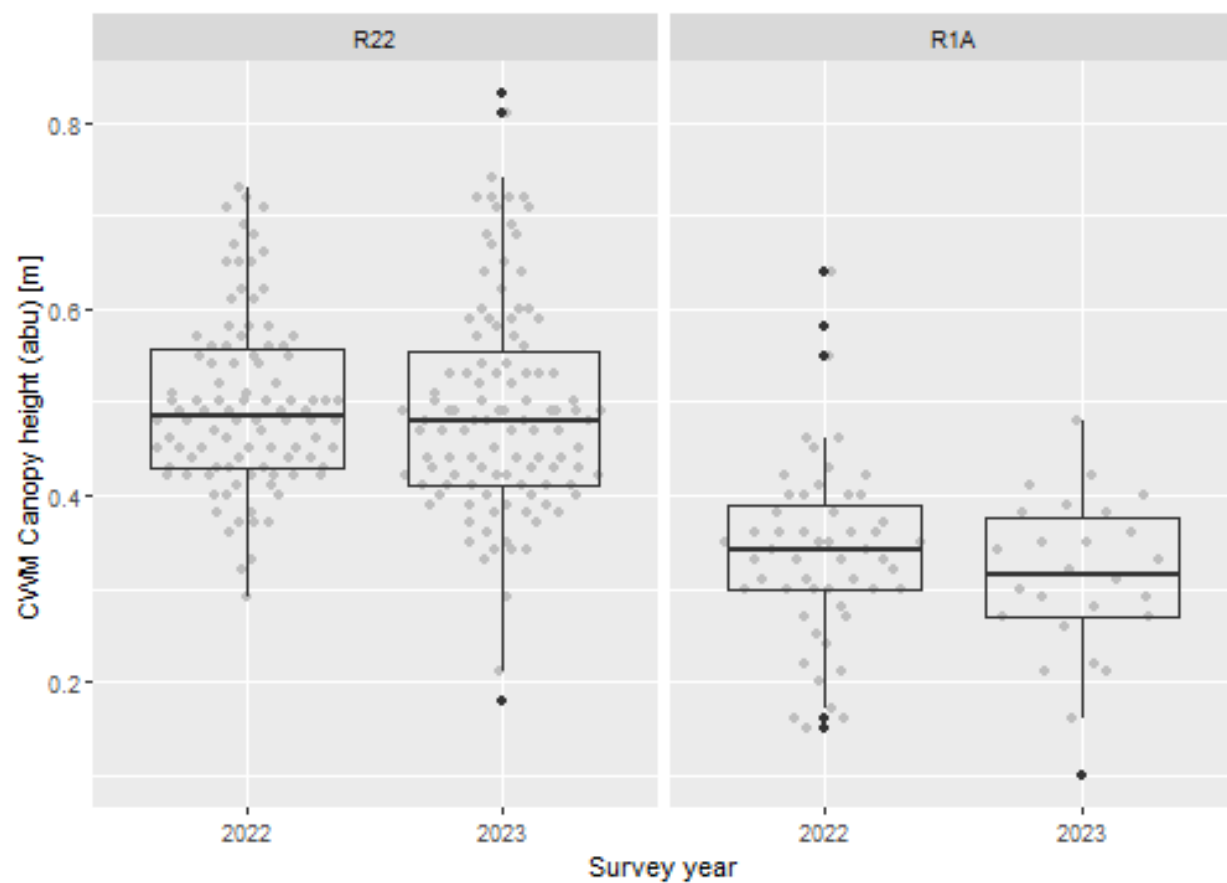
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3
##   esy4 site.type      n
##   <fct> <ord> <int>
## 1 R22 positive      25
## 2 R22 restored     167
## 3 R22 negative     16
## 4 R1A positive      20
## 5 R1A restored     52
## 6 R1A negative       9
```

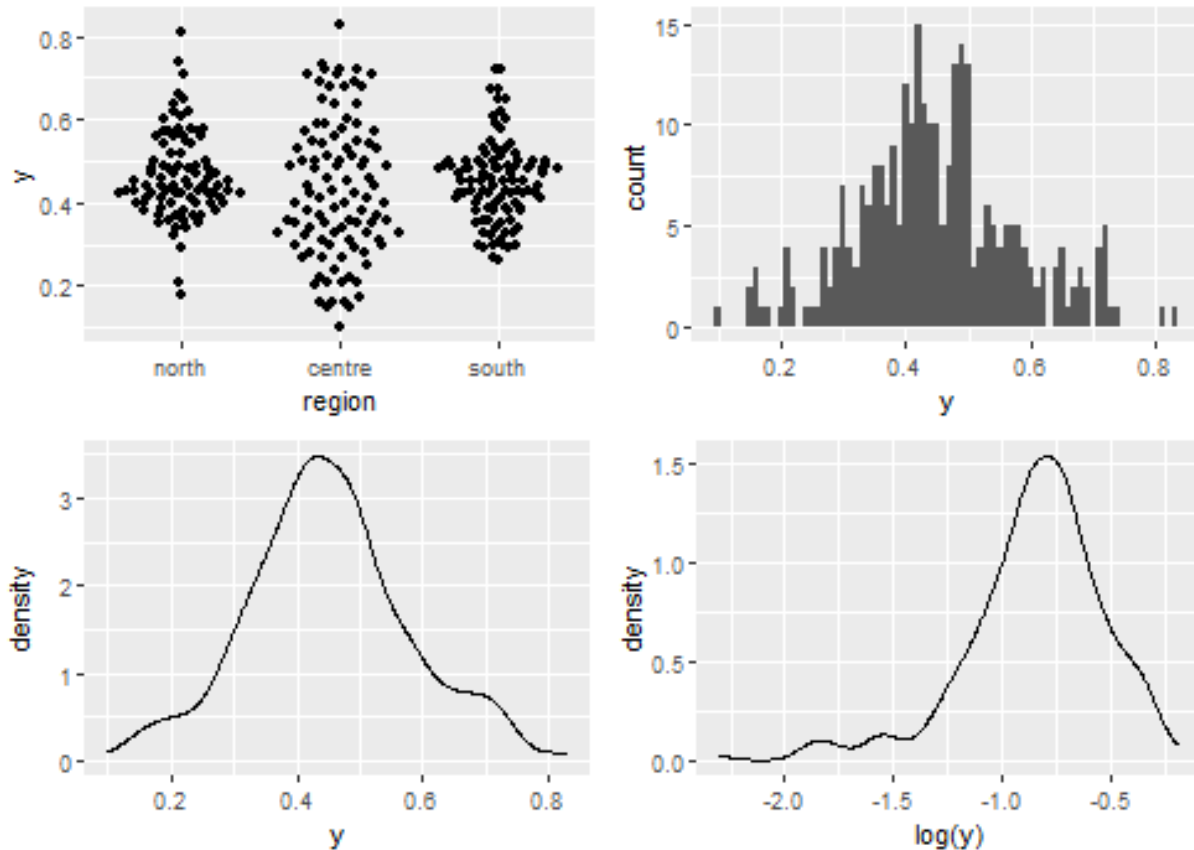
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum. ")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_height_esy4_cwm_1.Rdata"))
load(file = here("outputs", "models", "model_height_esy4_cwm_2.Rdata"))
m_1 <- m1
m_2 <- m2

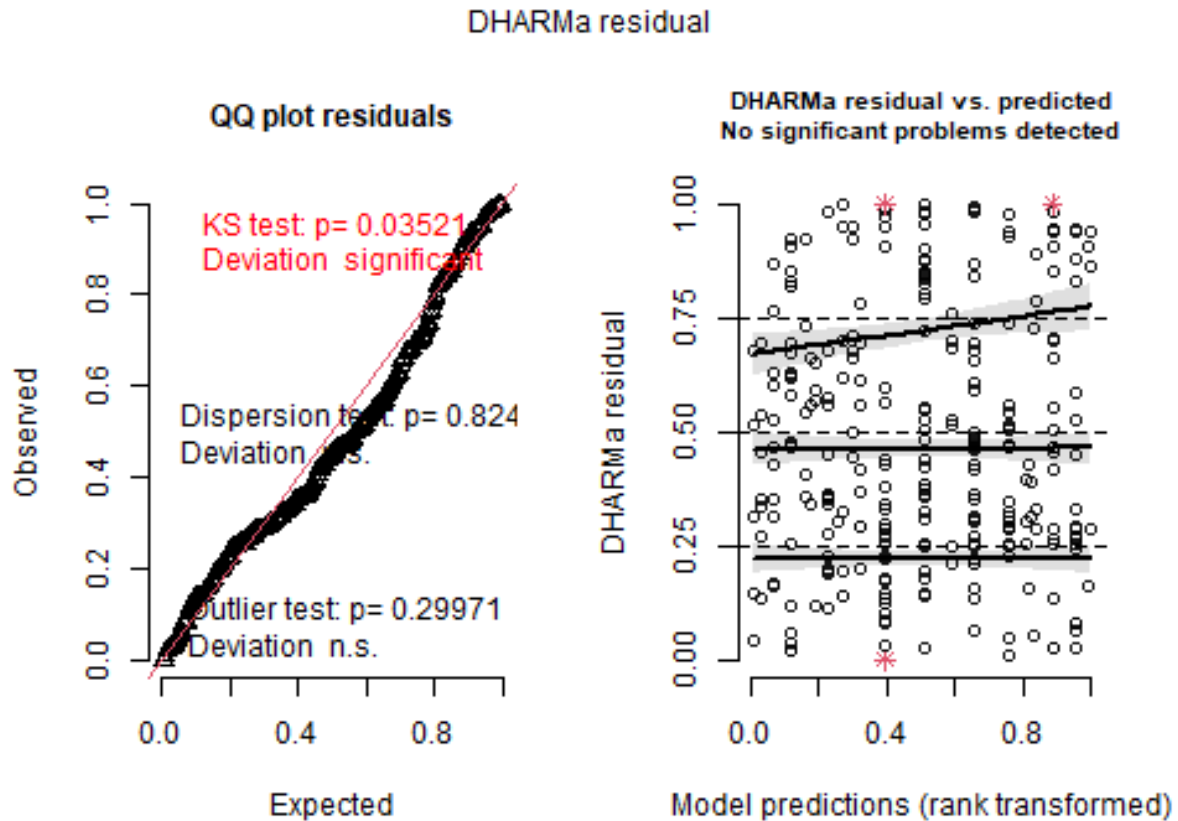
m_1@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + (1 |
##   id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = y ~ (esy4 + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

Model check

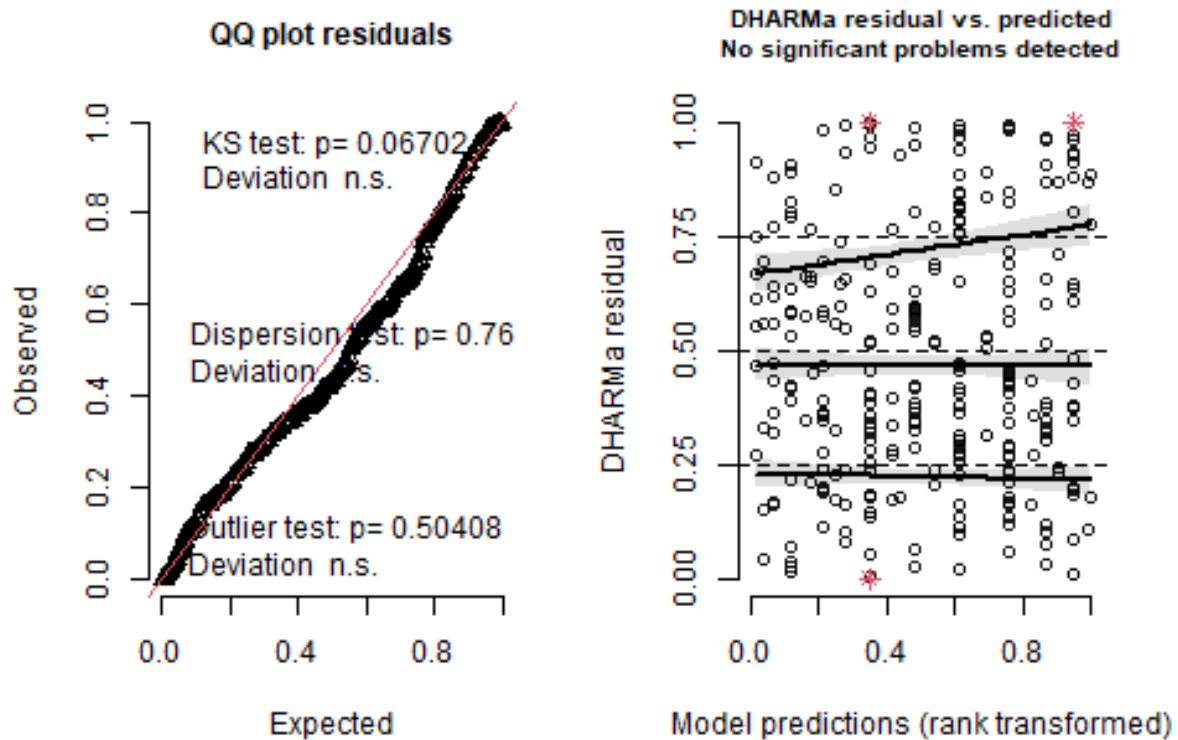
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

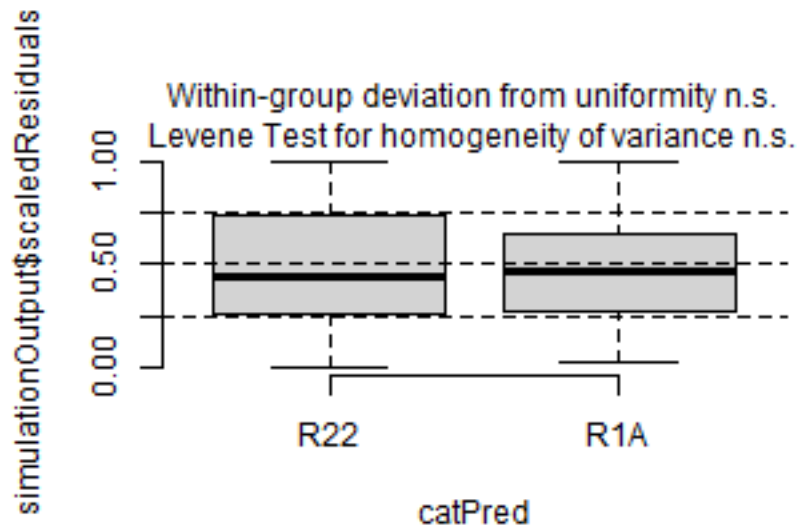


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

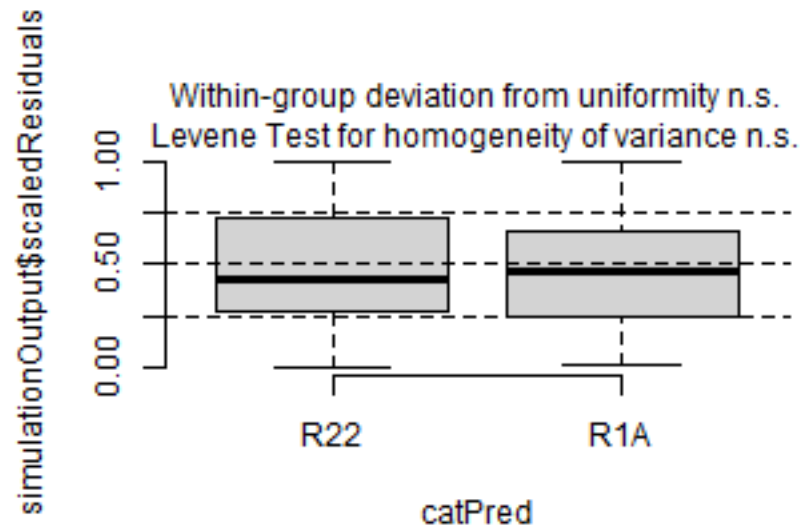

DHARMa residual



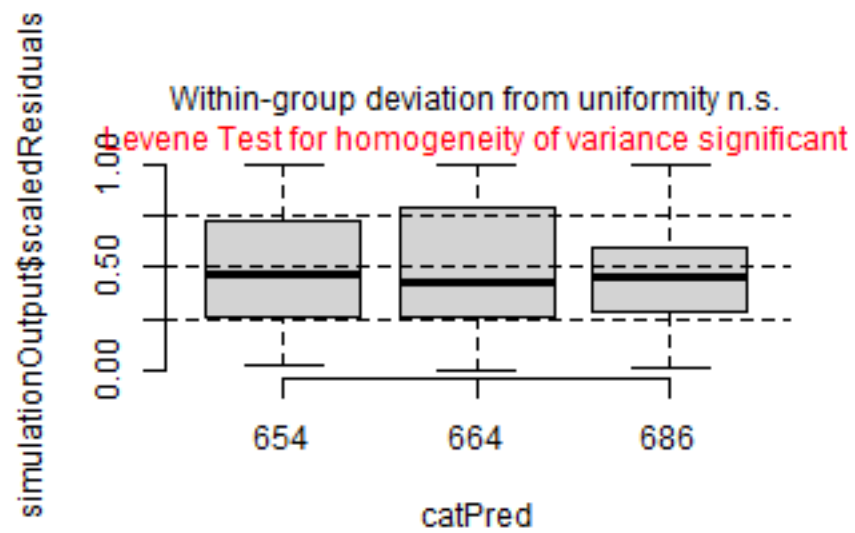
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



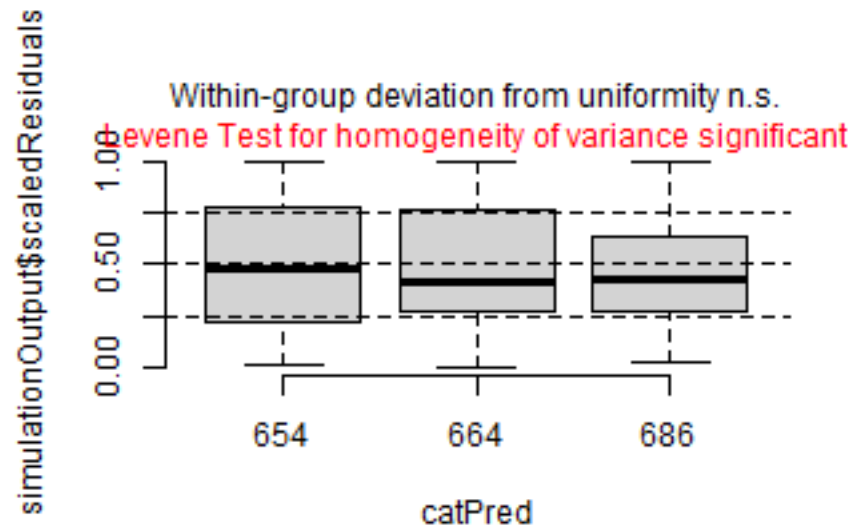
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



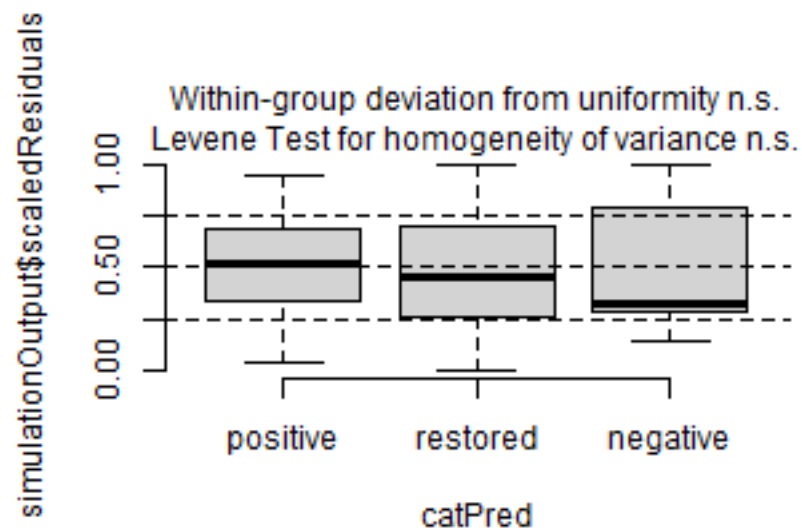
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



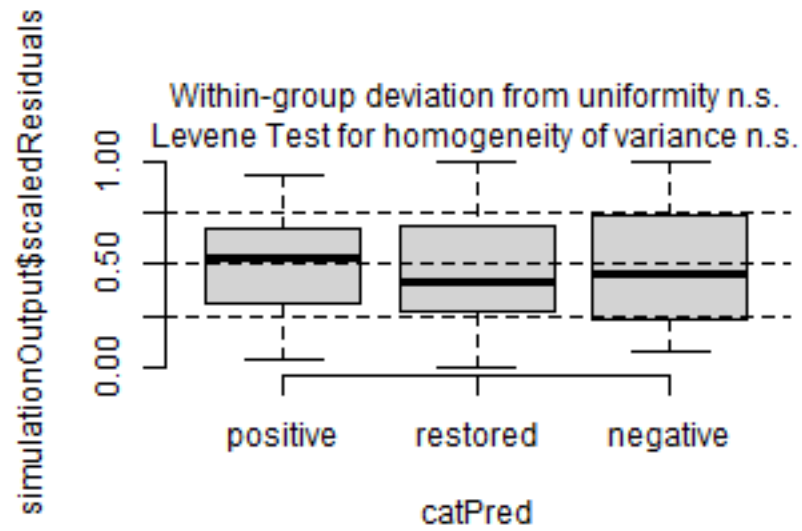
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



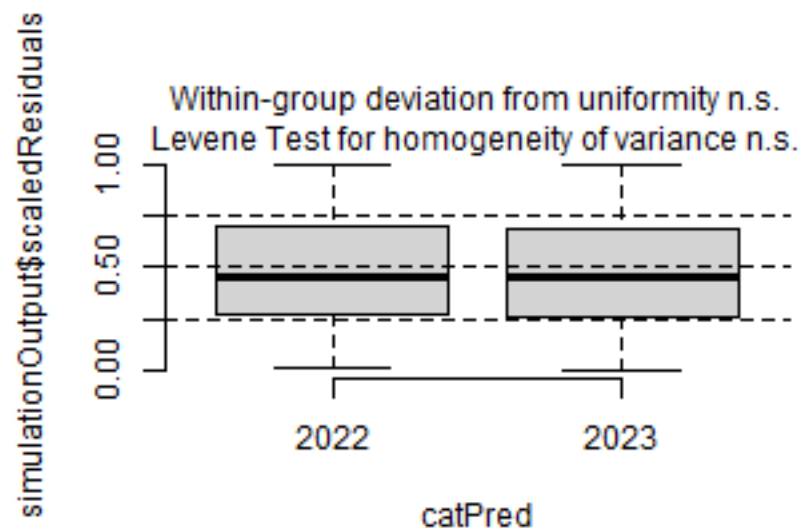
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



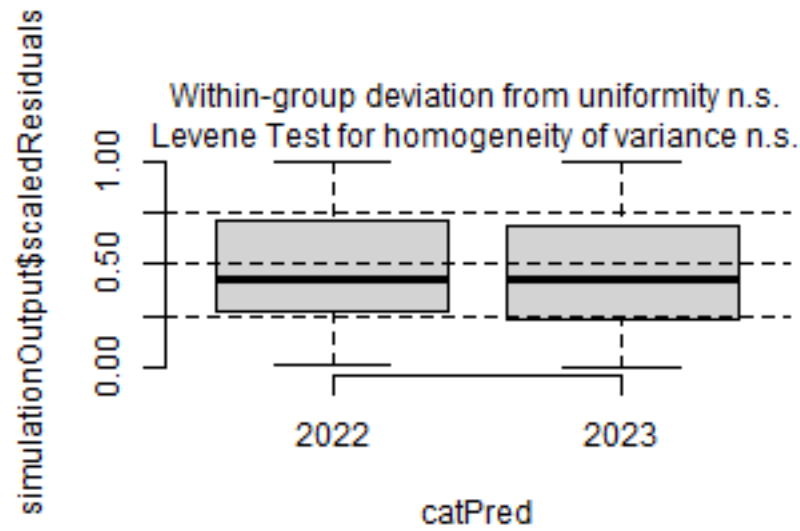
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



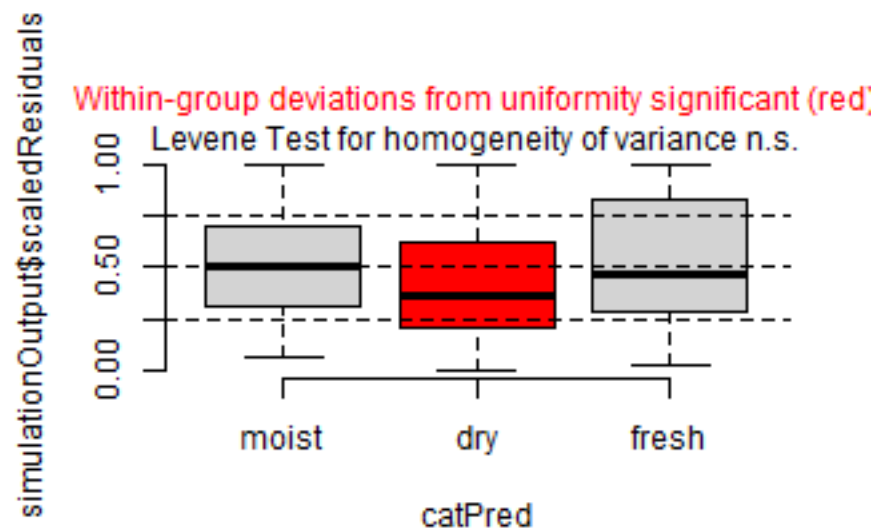
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



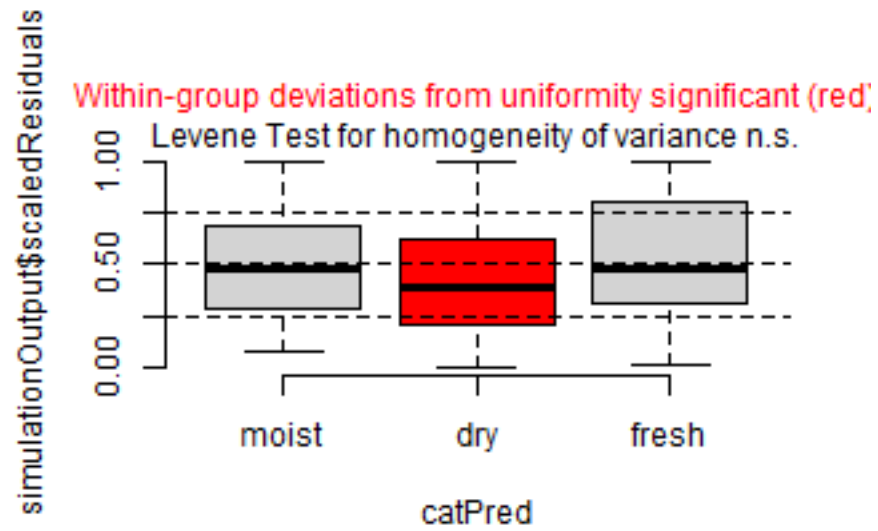
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



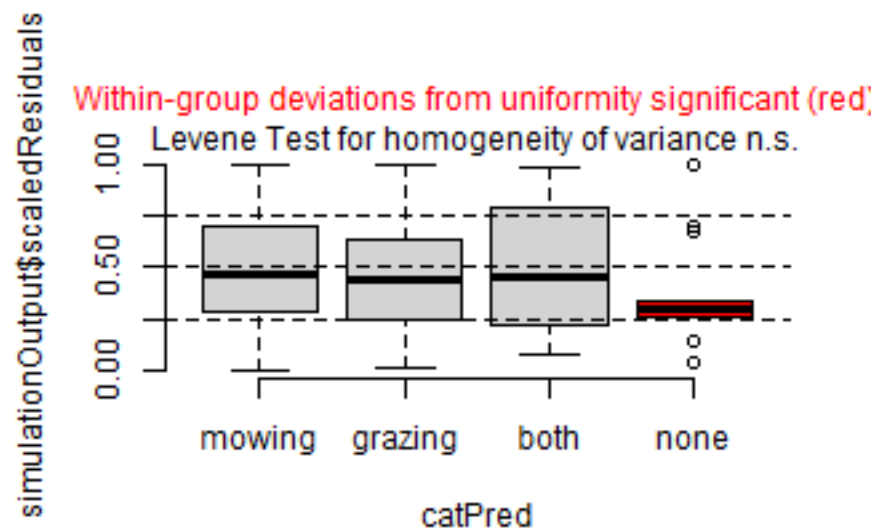
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



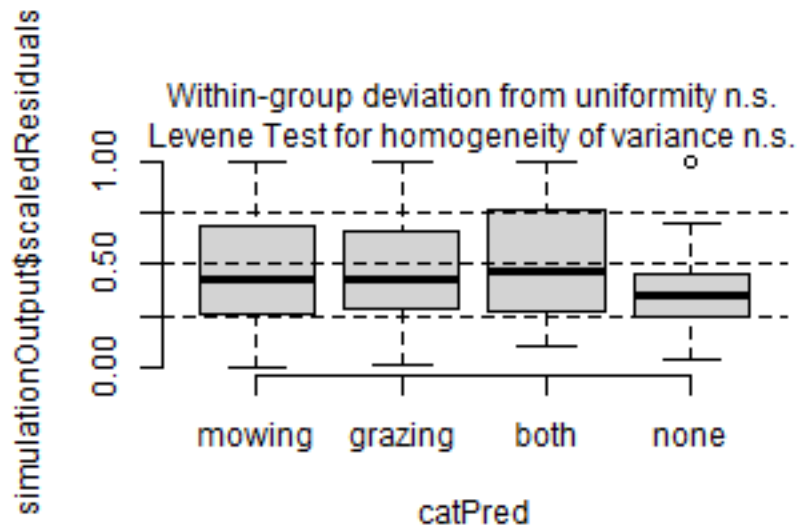
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove $VIF > 3$ or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          3.025606 1      1.739427
## site.type     1.976839 2      1.185749
## eco.id        1.769052 2      1.153281
## obs.year      1.046940 1      1.023201
## esy4:site.type 3.250758 2      1.342753
## esy4:eco.id    2.248951 1      1.499650
```

```
car::vif(m_2)
```

```
##              GVIF Df  GVIF^(1/(2*Df))
## esy4          5.540917 1      2.353915
## site.type     59.153304 2      2.773287
## eco.id        21.268885 2      2.147515
## obs.year      6.206095 1      2.491204
## esy4:site.type 9.398477 2      1.750912
## esy4:eco.id    2.585528 1      1.607958
## esy4:obs.year  2.998932 1      1.731743
## site.type:eco.id 77.800058 4      1.723346
## site.type:obs.year 9.600051 2      1.760226
## eco.id:obs.year 11.673695 2      1.848426
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##              R2m      R2c
## [1,] 0.4169676 0.715171
MuMIn::r.squaredGLMM(m_2)
```

```
##           R2m           R2c
## [1,] 0.4546629 0.7235414
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 12 -580.7186
## m_2 21 -570.2147
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##              Chisq Df Pr(>Chisq)
## (Intercept)   692.5248  1 < 2.2e-16 ***
## esy4           68.3004  1 < 2.2e-16 ***
## site.type       2.6087  2  0.2713428
## eco.id          15.6626  2  0.0003971 ***
## obs.year        0.2986  1  0.5847781
## esy4:site.type   1.5111  2  0.4697597
## esy4:eco.id      13.2930  1  0.0002664 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_1)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: y ~ esy4 * (site.type + eco.id) + obs.year + (1 | id.site)
## Data: sites
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##   -581.8   -537.9     302.9    -605.8      277
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.08092 -0.48018 -0.01625  0.41035  3.04610
##
## Random effects:
## Groups Name Variance Std.Dev.
## id.site (Intercept) 0.004589 0.06774
## Residual          0.004383 0.06620
## Number of obs: 289, groups: id.site, 118
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)    0.565453   0.021487  26.316
```



```

## esy4R1A          -0.230425    0.027882   -8.264
## site.type.L      0.038627    0.025364    1.523
## site.type.Q      0.012921    0.016537    0.781
## eco.id664        -0.087607    0.022400   -3.911
## eco.id686        -0.062701    0.022091   -2.838
## obs.year2023     -0.008427    0.015422   -0.546
## esy4R1A:site.type.L 0.055598    0.045928    1.211
## esy4R1A:site.type.Q 0.008598    0.030999    0.277
## esy4R1A:eco.id686  0.130702    0.035848    3.646
##
## Correlation of Fixed Effects:
##          (Intr) es4R1A st.t.L st.t.Q ec.664 ec.686 o.2023 e4R1A:...L
## esy4R1A      -0.641
## site.type.L -0.010 -0.042
## site.type.Q  0.447 -0.324  0.165
## eco.id664    -0.671  0.469  0.131 -0.063
## eco.id686    -0.632  0.458  0.056 -0.025  0.602
## obs.yer2023 -0.413  0.093  0.114 -0.018  0.073  0.003
## esy4R1A:...L -0.063  0.253 -0.511 -0.122 -0.041 -0.010  0.019
## esy4R1A:...Q -0.227  0.480 -0.119 -0.493  0.036  0.022  0.002  0.391
## es4R1A:.686  0.347 -0.506 -0.043  0.014 -0.341 -0.537  0.026  0.170
##          e4R1A:...Q
## esy4R1A
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## esy4R1A:...L
## esy4R1A:...Q
## es4R1A:.686  0.136
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

```

Forest plot

```

dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-0.2, 0.2) +
  theme_classic()

```

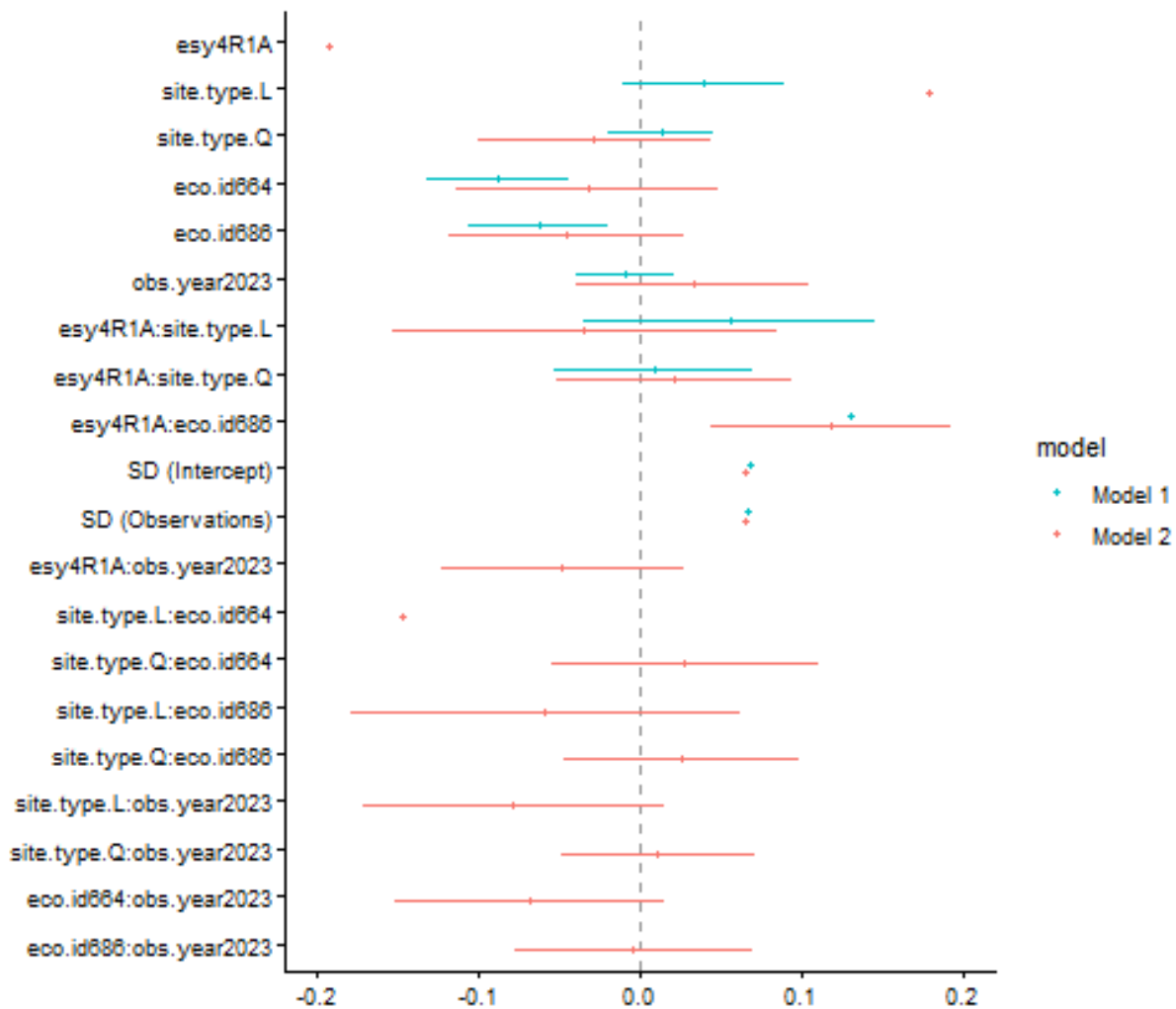
```

## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.

## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was

```

generated.



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_1,
  revpairwise ~ esy4,
  type = "response"
))
```

ESY EUNIS Habitat type

```
## $emmeans
## esy4 emmean SE df lower.CL upper.CL
## R22 0.511 0.0129 159 0.486 0.537
## R1A nonEst NA NA NA NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
```

```

## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast  estimate SE df z.ratio p.value
##   R1A - R22   nonEst NA NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
plot(emm, comparison = TRUE)

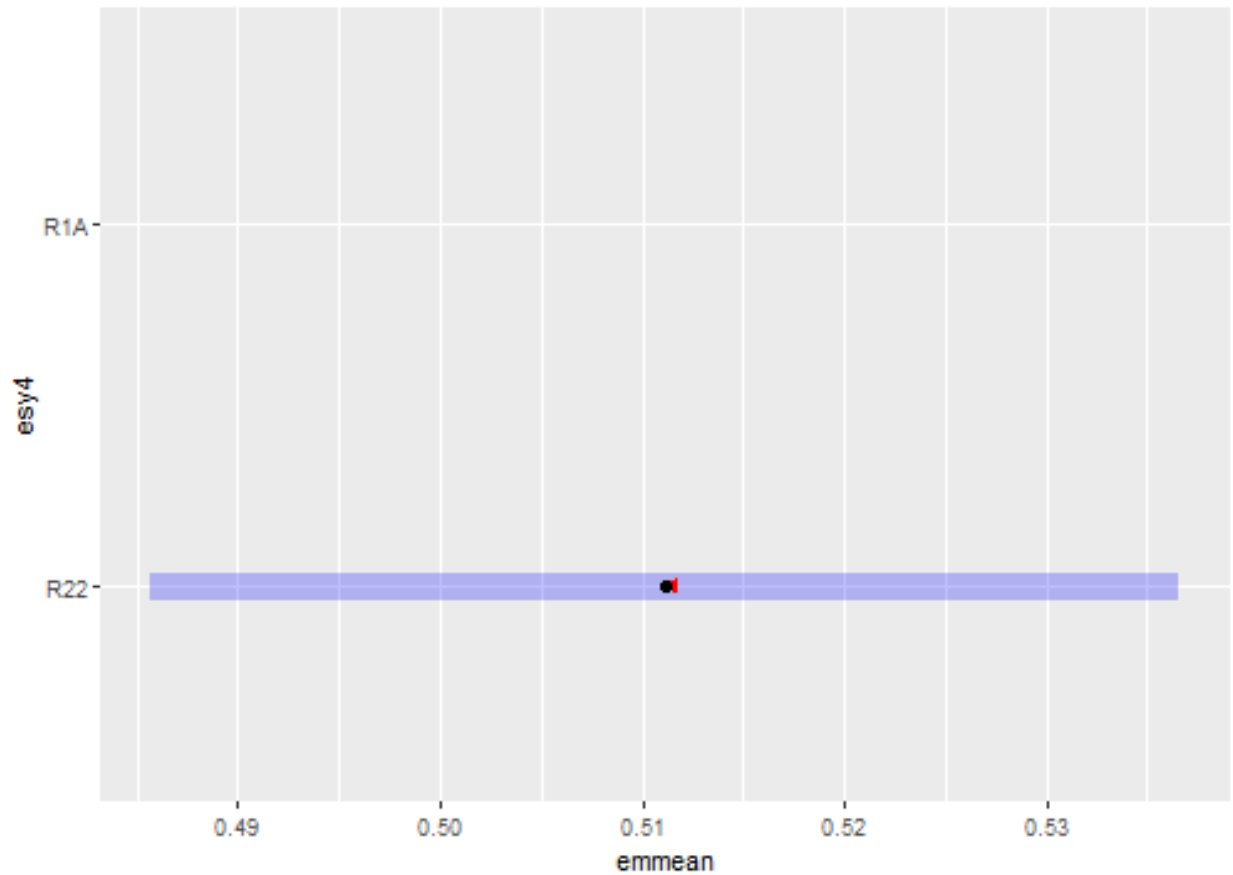
## Warning: `aes_()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`
## i The deprecated feature was likely used in the emmeans package.
##   Please report the issue at <https://github.com/rvlenth/emmeans/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

```



```
(emm <- emmeans(
  m_1,
  revpairwise ~ eco.id | esy4,
  type = "response"
))
```

Habitat type x Region

```
## $emmeans
## esy4 = R22:
##   eco.id emmean      SE df lower.CL upper.CL
##   654     0.561 0.0204 160    0.521    0.601
##   664     0.474 0.0170 130    0.440    0.507
##   686     0.499 0.0172 162    0.465    0.532
##
## esy4 = R1A:
##   eco.id emmean      SE df lower.CL upper.CL
##   654     0.331 0.0217 129    0.288    0.374
##   664   nonEst     NA  NA      NA      NA
##   686     0.399 0.0305 158    0.339    0.459
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
```

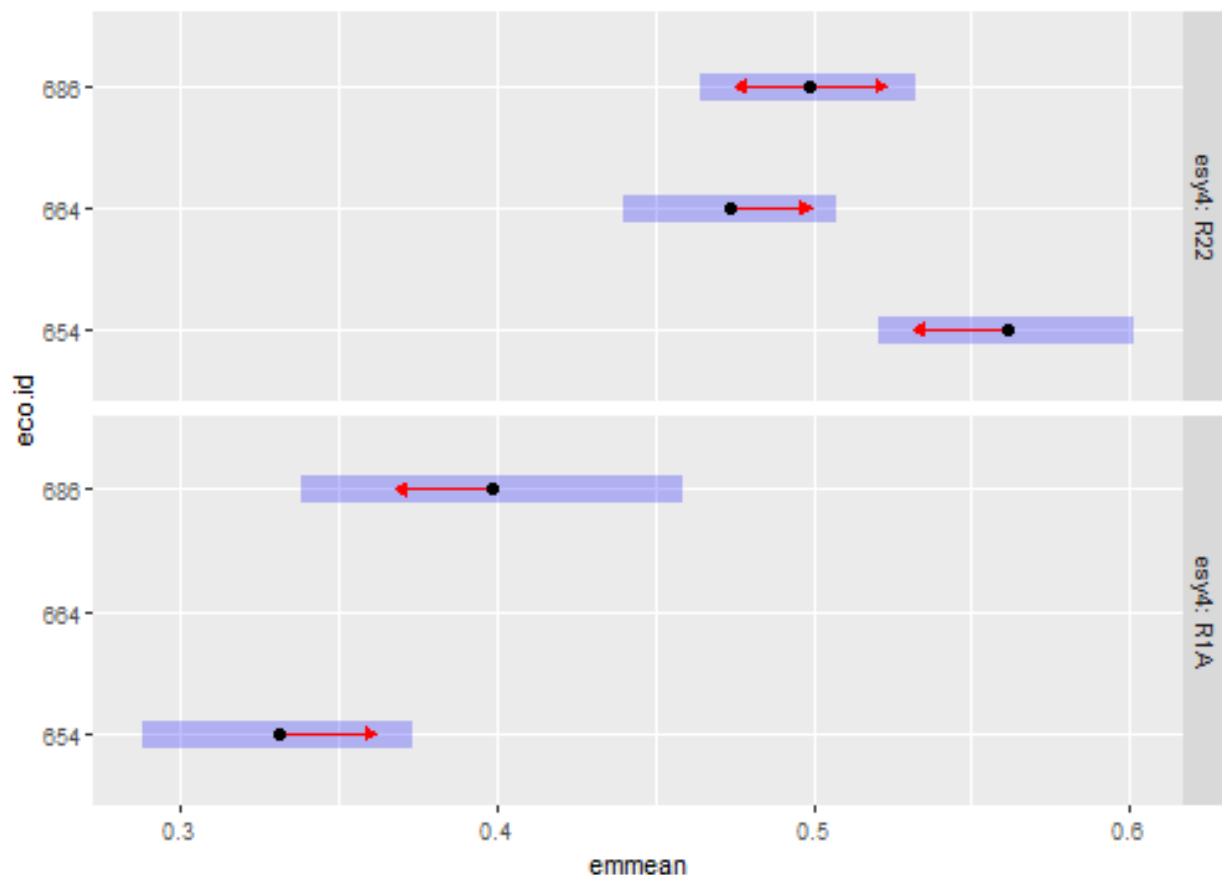
```
## $contrasts
## esy4 = R22:
## contrast          estimate      SE  df t.ratio p.value
## eco.id664 - eco.id654 -0.0876 0.0233 140 -3.756 0.0007
## eco.id686 - eco.id654 -0.0627 0.0230 154 -2.730 0.0193
## eco.id686 - eco.id664  0.0249 0.0207 137  1.205 0.4520
##
## esy4 = R1A:
## contrast          estimate      SE  df t.ratio p.value
## eco.id664 - eco.id654 nonEst     NA  NA      NA      NA
## eco.id686 - eco.id654  0.0680 0.0316 165  2.149 0.0331
## eco.id686 - eco.id664 nonEst     NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
```

```
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
  m_1,
  revpairwise ~ site.type | esy4,
  type = "response"
))
```

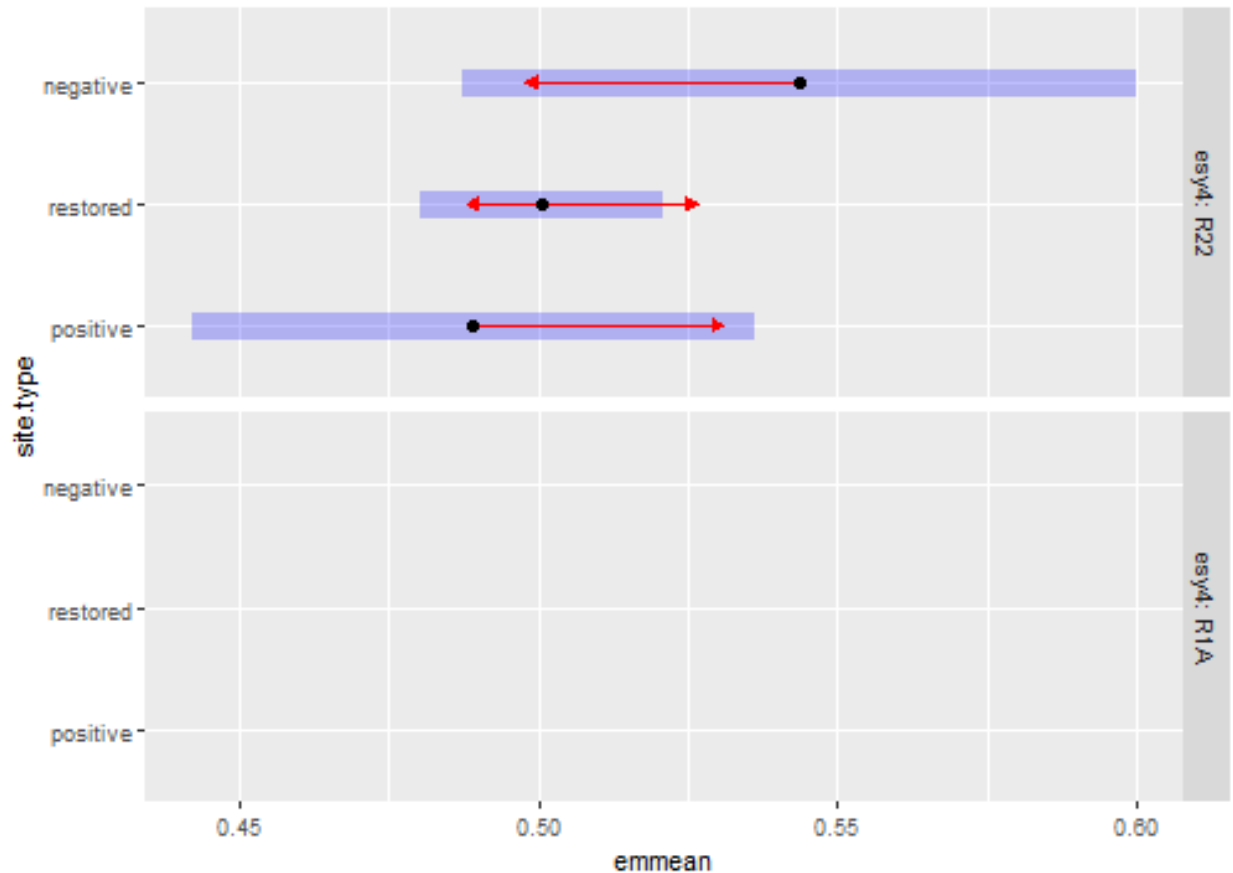
Habitat type x Site type

```
## $emmeans
## esy4 = R22:
##   site.type emmean      SE  df lower.CL upper.CL
##   positive   0.489 0.0238 173    0.442    0.536
##   restored   0.501 0.0103 134    0.480    0.521
##   negative   0.544 0.0285 152    0.487    0.600
##
## esy4 = R1A:
##   site.type emmean      SE  df lower.CL upper.CL
##   positive nonEst     NA  NA      NA      NA
##   restored nonEst     NA  NA      NA      NA
##   negative nonEst     NA  NA      NA      NA
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## esy4 = R22:
##   contrast          estimate      SE  df t.ratio p.value
##   restored - positive    0.0115 0.0257 164    0.447  0.8958
##   negative - positive    0.0546 0.0372 160    1.467  0.3094
##   negative - restored    0.0431 0.0303 150    1.424  0.3313
##
## esy4 = R1A:
##   contrast          estimate      SE  df t.ratio p.value
##   restored - positive    0.0403 0.0347 160    1.159  0.4794
##   negative - positive    0.1333 0.0584 126    2.283  0.0618
##   negative - restored    0.0930 0.0536 120    1.734  0.1969
##
## Results are averaged over the levels of: eco.id, obs.year
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_segment()`).
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMa_0.4.7    patchwork_1.3.0 ggbeeswarm_0.7.2
## [5] lubridate_1.9.4 forcats_1.0.0    stringr_1.5.2   dplyr_1.1.4
## [9] purrr_1.1.0     readr_2.1.5     tidyr_1.3.1     tibble_3.3.0
```

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## [13] ggplot2_4.0.0    tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6       magrittr_2.0.4     multcomp_1.4-28
## [7] compiler_4.5.2    mgcv_1.9-3         vctrs_0.6.5
## [10] pkgconfig_2.0.3   crayon_1.5.3       fastmap_1.2.0
## [13] backports_1.5.0   labeling_0.4.3     utf8_1.2.6
## [16] ggstance_0.3.7    promises_1.3.2     rmarkdown_2.30
## [19] tzdb_0.5.0        nloptr_2.2.1       bit_4.6.0
## [22] xfun_0.53         later_1.4.2        broom_1.0.8
## [25] parallel_4.5.2    R6_2.6.1           gap.datasets_0.0.6
## [28] stringi_1.8.7     qgam_2.0.0         RColorBrewer_1.1-3
## [31] car_3.1-3         boot_1.3-32        estimability_1.5.1
## [34] Rcpp_1.1.0        iterators_1.0.14    knitr_1.50
## [37] zoo_1.8-14        parameters_0.25.0   httpuv_1.6.16
## [40] Matrix_1.7-4      splines_4.5.2       timechange_0.3.0
## [43] tidysselect_1.2.1 rstudioapi_0.17.1   abind_1.4-8
## [46] yaml_2.3.10       MuMIn_1.48.11       doParallel_1.0.17
## [49] codetools_0.2-20  lattice_0.22-7      plyr_1.8.9
## [52] bayestestR_0.15.3 shiny_1.10.0        withr_3.0.2
## [55] S7_0.2.0          evaluate_1.0.5      marginaleffects_0.25.1
## [58] survival_3.8-3    pillar_1.11.1       gap_1.6
## [61] carData_3.0-5     foreach_1.5.2       stats4_4.5.2
## [64] reformulas_0.4.1  insight_1.2.0       generics_0.1.4
## [67] vroom_1.6.6       rprojroot_2.0.4     hms_1.1.3
## [70] scales_1.4.0      minqa_1.2.8         xtable_1.8-4
## [73] glue_1.8.0        tools_4.5.2         data.table_1.17.8
## [76] lme4_1.1-37       mvtnorm_1.3-3       grid_4.5.2
## [79] rbibutils_2.3     datawizard_1.1.0    nlme_3.1-168
## [82] Rmisc_1.5.1       performance_0.13.0  beeswarm_0.4.0
## [85] vipor_0.4.7       Formula_1.2-5       cli_3.6.5
## [88] gtable_0.3.6      digest_0.6.37       pbkrtest_0.5.4
## [91] TH.data_1.1-3     farver_2.1.2        htmltools_0.5.8.1
## [94] lifecycle_1.0.4   mime_0.13           bit64_4.6.0-1
## [97] dotwhisker_0.8.4  MASS_7.3-65

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